

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	
Project Name	Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development. The codebase reviewed includes the data preprocessing and feature engineering scripts, the model training notebooks for Logistic Regression, Decision Tree, Random Forest, and XGBoost, and the Flask web application integrated with the IBM Watson Machine Learning deployment.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP-8 4-space indentation followed consistently across preprocessing scripts, model training notebooks, and Flask app.py.
2	Proper File Structure	Files and folders are logically organized	Yes	Project separated into /data, /notebooks, /model, /templates, /static, and app.py at root, following standard Flask conventions.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as income_type, employment_years, credit_score, and approval_status are self-descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like preprocess_input(), train_model(), evaluate_model(), and predict_approval() clearly convey purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Preprocessing and model training notebooks are well commented; some helper functions in app.py need additional inline comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Feature engineering, model training, and prediction logic are separated into distinct, importable modules.

7	No Redundant Code	Duplicate or unused code is removed	Partial	Minor duplication exists between the EDA notebook and preprocessing script; recommended to consolidate into a shared utility module.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Flask routes use try/except blocks to catch invalid form inputs and model/API errors, returning user-friendly messages.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	data_preprocessing.py (feature engineering & encoding functions)	Python (pandas, scikit-learn)	Reused across EDA notebook, model training scripts, and Flask prediction pipeline	High
2	model_utils.py (train_model, evaluate_model functions)	Python (scikit-learn, XGBoost)	Reused for training and evaluating all four classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost)	High
3	predict_approval() prediction function	Python (Flask, pickle)	Reused in Flask app.py route and IBM Watson deployment wrapper script	High
4	form_input.html (Bootstrap web form template)	HTML / CSS / Bootstrap	Reused for both the analyst-facing screening page and the customer self-service eligibility page	Medium
5	watson_deploy.py (IBM Watson ML deployment script)	Python (ibm-watson-machine-learning SDK)	Reused to deploy and update the selected best-performing model on IBM Cloud	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean, logically organized project structure with clear separation of data, model, and web app layers.
Readability	4	Descriptive naming and mostly sufficient comments; a few utility functions in app.py could use more documentation.
Reusability	5	Preprocessing and model training functions are modular and reused consistently across notebooks and the Flask app.
Documentation / Comments	4	Notebooks are well documented; a project-level README with setup and deployment steps would further improve onboarding.
Overall Score	4.5	Well-structured, readable, and reusable codebase suitable for maintenance and future extension.